

Apply ratio, proportion and rates of change

Solve problems using ratio, proportion and rates in real-life contexts.

Step 55

★ Today I am learning to apply ratio, proportion and rates of change.

1. Simplify Ratios

Write each ratio in simplest form.

a) $6 : 9 =$ _____

b) $12 : 18 =$ _____

c) $15 : 25 =$ _____

d) $20 : 30 =$ _____

e) $24 : 36 : 48 =$ _____

f) $9 : 12 : 15 =$ _____

g) $14 : 21 : 28 =$ _____

h) $16 : 24 : 40 =$ _____

Think about it!

- Divide all parts of the ratio by the highest common factor.
- Ratios show how quantities compare.

2. Share in a Given Ratio

Share each amount in the given ratio.

a) Share £60 in the ratio 2 : 3.
 £ _____ and £ _____

b) Share £84 in the ratio 3 : 5.
 £ _____ and £ _____

c) Share 90 in the ratio 1 : 2 : 3.
 _____, _____ and _____

d) Share £120 in the ratio 2 : 3 : 5.
 £ _____, £ _____ and £ _____

e) Share 72 in the ratio 2 : 5 : 7.
 _____, _____ and _____

f) Share £150 in the ratio 1 : 4 : 5.
 £ _____, £ _____ and £ _____

Remember:

- Add the parts of the ratio.
- Work out the value of one part.
- Multiply each part by that value.

3. Direct Proportion

Solve each problem.

a) y is directly proportional to x .
 When $x = 4$, $y = 18$.
 Find y when $x = 7$.
 Answer: _____

b) y is directly proportional to x .
 When $x = 5$, $y = 35$.
 Find y when $x = 12$.
 Answer: _____

c) y is directly proportional to x .
 When $x = 6$, $y = 24$.
 Find y when $x = 15$.
 Answer: _____

d) y is directly proportional to x .
 When $x = 8$, $y = 56$.
 Find x when $y = 98$.
 Answer: _____

e) y is directly proportional to x .
 When $x = 3$, $y = 27$.
 Find x when $y = 63$.
 Answer: _____

f) y is directly proportional to x .
 When $x = 7$, $y = 42$.
 Find y when $x = 20$.
 Answer: _____

Think about it!

- In direct proportion:

$$y = kx$$
- Find k using the given pair of values.
- Then use k to find the unknown.

4. Inverse Proportion

Solve each problem.

a) y is inversely proportional to x .
 When $x = 4$, $y = 24$.
 Find y when $x = 6$.
 Answer: _____

b) y is inversely proportional to x .
 When $x = 5$, $y = 30$.
 Find y when $x = 10$.
 Answer: _____

c) y is inversely proportional to x .
 When $x = 8$, $y = 40$.
 Find y when $x = 16$.
 Answer: _____

d) y is inversely proportional to x .
 When $x = 3$, $y = 27$.
 Find x when $y = 9$.
 Answer: _____

e) y is inversely proportional to x .
 When $x = 6$, $y = 18$.
 Find x when $y = 12$.
 Answer: _____

f) y is inversely proportional to x .
 When $x = 9$, $y = 36$.
 Find x when $y = 8$.
 Answer: _____

Remember:

- In inverse proportion:

$$y = \frac{k}{x}$$
- Find k using the given pair of values.
- Then use k to find the unknown.

5. Proportion (Using Fractions)

Solve each proportion.

a) $\frac{2}{3} = \frac{x}{12}$

d) $\frac{3}{8} = \frac{3}{x}$

c) $\frac{4}{7} = \frac{x}{21}$

e) $\frac{5}{x} = \frac{15}{24}$

b) $\frac{3}{5} = \frac{9}{x}$

d) $\frac{x}{86} = \frac{3}{16}$

f) $\frac{7}{9} = \frac{14}{x}$

Answer: _____

Think about it!

- The cross products are equal.
- Use $\frac{a}{b} = \frac{c}{d} \Rightarrow ad = bc$.

6. Rates of Change

Solve each problem.

a) A car travels 240 km in 3 hours.
 What is the speed in km/h?
 Answer: _____

b) A recipe uses 120 g of sugar for 8 cakes.
 How much sugar is needed for 15 cakes?
 Answer: _____

c) A tank is filled with 300 litres of water in 6 minutes.
 How many litres are filled in 10 minutes?
 Answer: _____

d) A cyclist travels 90 km in 2.5 hours.
 What is the average speed in km/h?
 Answer: _____

e) A machine produces 450 items in 6 hours.
 How many items will it produce in 9 hours?
 Answer: _____

f) A car uses 7.5 litres of fuel to travel 105 km.
 How far can it travel with 12 litres of fuel?
 Answer: _____

Remember:

- Rate = amount $\div$ time
- Keep the units the same.
- For direct proportion, use ratios.

7. Real-World Problems

Solve each problem.

a) The ratio of boys to girls in a class is 3 : 5.
 There are 24 boys. How many girls are there?
 Answer: _____

b) A map scale is 1 cm : 5 km.
 If the distance between two towns is 15 cm on the map, what is the actual distance in km?
 Answer: _____

c) The number of workers and the time taken to complete a job are inversely proportional. If 6 workers take 10 days to complete the job, how long will 4 workers take?
 Answer: _____

d) A population of bacteria doubles every 3 hours. If the initial population is 500, what will it be after 12 hours?
 Answer: _____

Think about it!

- Read carefully.
- Identify the type of relationship.
- Use ratios, proportions or rates to solve.

My Learning Check

Circle how confident you feel.



I can do it!



Almost there!



I need more practice.



I need help.

Parent Observation (optional)

Please tick the boxes that apply.

- ☐ Understands the concept
☐ Uses strategies confidently
☐ Solves problems accurately

- ☐ Explains thinking clearly
☐ Needs more practice

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____